Experiencing Immersive Virtual Nature for Well-Being, Restoration, Performance, and Nature Connectedness: A Scoping Review

Jeewoo Kim, Svara Patel, Hyeongil Nam, Janghee Cho, and Kangsoo Kim

Abstract—This paper presents a scoping review of immersive virtual nature experiences delivered via head-mounted displays (HMDs) and their role in promoting well-being, psychological restoration, cognitive performance, and nature connectedness. As access to natural environments becomes increasingly constrained by urbanization, technological lifestyles, and environmental change, immersive technologies offer a scalable and accessible alternative for engaging with nature. Guided by three core research questions, this review explores how HMD-mediated immersive technologies have been used to promote nature connectedness and well-being, what trends and outcomes have been observed across applications, and what methodological gaps or limitations exist in this growing body of work. Fifty-five peer-reviewed studies were analyzed and categorized into six key implication areas: emotional well-being, stress reduction, cognitive performance, attention recovery, restorative benefits, and nature connectedness. The review identifies immersive virtual nature as a promising application of extended reality (XR) technologies, with potential across healthcare, education, and daily life, while also emphasizing the need for more consistent methodologies and long-term research.

Index Terms—Scoping review, augmented reality, virtual reality, virtual nature, well-being, restoration, cognitive performance, nature connectedness

1 Introduction

Access to natural environments has long been associated with a range of psychological, cognitive, and physiological benefits. A growing body of empirical research demonstrates that exposure to nature supports mental well-being, facilitates emotional regulation, restores depleted cognitive resources, and enhances one's sense of connection to the natural world [5, 30, 61]. However, in an increasingly urbanized and technologically mediated world, direct access to restorative natural settings is becoming limited for many individuals. Urban living, mobility constraints, institutional settings (e.g., hospitals, long-term care), and climate-related disruptions often reduce opportunities for meaningful nature contact. This has led to what some researchers have described as a "nature deficit" [9], which is now recognized as a contributing factor to the growing global burden of mental health challenges.

In response to this gap, researchers and practitioners have turned to *virtual nature experiences*—digitally mediated representations of natural environments delivered via virtual reality (VR), augmented reality (AR), 360° video, or immersive simulations—as a potential alternative or supplement to real-world nature exposure. Preliminary evidence suggests that virtual nature can elicit similar affective, cognitive, and restorative outcomes as physical nature, including reductions in stress and anxiety, improved mood, enhanced attention, and increased feelings of connectedness to nature [7, 19, 42]. Such interventions are gaining popularity in clinical, educational, and workplace contexts due to their scalability, customizability, and accessibility.

Despite this growing interest, the evidence base for virtual nature remains fragmented. Studies vary widely in their design, population, duration, technological platforms, and outcome measures. Additionally, little is known about the mechanisms by which virtual nature exerts its effects, how it compares to real nature exposure, or the extent to which it fosters long-term improvements in well-being and nature connectedness. These limitations hinder the development of evidence-based guidelines for the effective design and deployment of virtual nature interventions.

Given the rising demand for scalable mental health tools, the rapid adoption of immersive technologies, and the urgent need to reconnect individuals with nature in the face of global ecological crises, a comprehensive synthesis of the current literature is both timely and essential. This scoping review aims to examine the state of empirical research on virtual nature experiences, with a specific focus on their effects on (1) psychological well-being, (2) cognitive and emotional restoration, (3) nature connectedness, and (4) other health/performance-related outcomes. By identifying key findings, methodological trends, and research gaps, this review seeks to provide a foundation for future research, guide best practices in intervention design, and inform cross-disciplinary applications of virtual nature in healthcare, education, and environmental communication.

To systematically address existing gaps and variations in the literature, studies are categorized according to immersive technology types (VR and AR) and sensory modalities, recognizing their impact on user engagement and outcomes. Additionally, practical applications and key study characteristics—including participant demographics, measures used, and study durations—are analyzed to understand the meaning of user study in papers. Through this comprehensive literature review, we conduct a detailed analysis aligned with our specific focus, aiming to highlight gaps in current research and uncover valuable insights.

This review particularly seeks to address the following research questions (RQs):

- **RQ1:** How have immersive technologies been used to promote well-being and strengthen nature connectedness, and what future advancements could further enhance their effectiveness?
- **RQ2:** What trends and outcomes have emerged across different thematic applications of immersive nature experiences for well-being?
- **RQ3:** What methodological limitations and research gaps exist in evaluating virtual nature's effects on well-being and nature connectedness?

To contextualize these questions and clarify the contribution of this review, the following section outlines existing research on nature connectedness, immersive technologies, and their intersection within well-being literature.

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2 Related Work

As interest in virtual nature interventions grows, a range of reviews have explored their psychological, cognitive, and physiological effects. These reviews highlight how immersive VR and virtual landscapes can support general well-being [21], positive affect [43], and emotional recovery [38], with some focusing on specific features such as natural sounds, lighting, and water elements that contribute to stress relief [24]. The immersive capabilities of VR have made nature-based approaches such as forest therapy a frequently used method for stress reduction, reflecting a broader trend toward simulating restorative natural environments in virtual settings [36, 54]. In healthcare settings, virtual nature has demonstrated modest therapeutic effects, such as reducing anxiety and pain [26]. Additionally, virtual nature has shown potential in mindfulness-compassion programs for older adults, contributing to improved mental health and well-being [67]. However, despite these positive outcomes, findings on virtual nature's overall effectiveness remain mixed, and further research is necessary to optimize its applications. While natural environments are well-documented for their stress-reducing effects [22], virtual nature's impact is still being explored, with additional challenges such as refining intervention parameters and understanding long-term benefits [74].

Beyond its direct effects on well-being, virtual nature is grounded in two key theoretical perspectives: the biophilia hypothesis and Attention Restoration Theory (ART). "Biophilia," introduced by Edward O. Wilson [83], describes the innate human tendency to connect with nature and living systems, while Attention Restoration Theory (ART) [34] posits that exposure to natural environments helps restore directed attention and reduce mental fatigue. Immersive VR offers cost-effective, flexible, and realistic simulations of indoor spaces, enabling evaluation of the psychological and cognitive effects of various biophilic features on occupants [16]. VR environments can replicate biophilic patterns and assess human responses to virtual nature, offering insights into how these designs influence well-being. Integrating biophilic design into VR can also enrich built environments, supporting stress management and occupatioal therapy for individuals with Parkinson's disease [31] However, challenges such as cybersickness and limited sensory input still hinder VR's full potential in biophilic design [55], underscoring the need for further research to optimize virtual nature experiences and address technological barriers.

Virtual nature also holds potential for fostering nature connectedness, a psychological bond linked to well-being. Research has examined how digital technologies, including HCI and AR, support nature engagement, with studies suggesting positive effects on human-nature relationships [13, 81]. Meta-analyses of various interventions indicate consistent short-term increases in nature connectedness [6, 72]. Despite this, a review of nature-centric technologies highlights the need for more intentional design to create deeper connections with nature [76]. In contrast to prior reviews, this paper focuses on immersive virtual nature using HMDs and synthesizes evidence across psychological outcomes, cognitive effects, and nature connectedness to inform future work. As such, it complements existing work by offering a more integrative perspective to guide future immersive nature applications.

3 Methodology

In this research, a scoping literature review was conducted in accordance with the PRISMA guidelines [62]. The initial search was carried out on January 28, 2025, across five databases: Compendex (236 records), the University Library (100 records), PubMed (16 records), IEEE Xplore (42 records), and the ACM Digital Library (90 records), resulting in a total of 484 records. These databases were selected to provide broad and interdisciplinary coverage of research specifically focused on immersive virtual nature experiences across a variety of fields. The search strategy combined three key themes—nature connectedness, immersive technologies, and health or well-being outcomes—using the following search terms: ("nature connectedness" OR "nature relatedness" OR "connection with nature" OR "biophilia hypothesis" OR "biophilic") AND ("virtual reality" OR "augmented reality" OR "mixed reality" OR "extended reality") AND ("well being" OR "well-being" OR "wellbeing" OR "wellness" OR "health" OR "therapy" OR "therapeutic" OR "treatment").

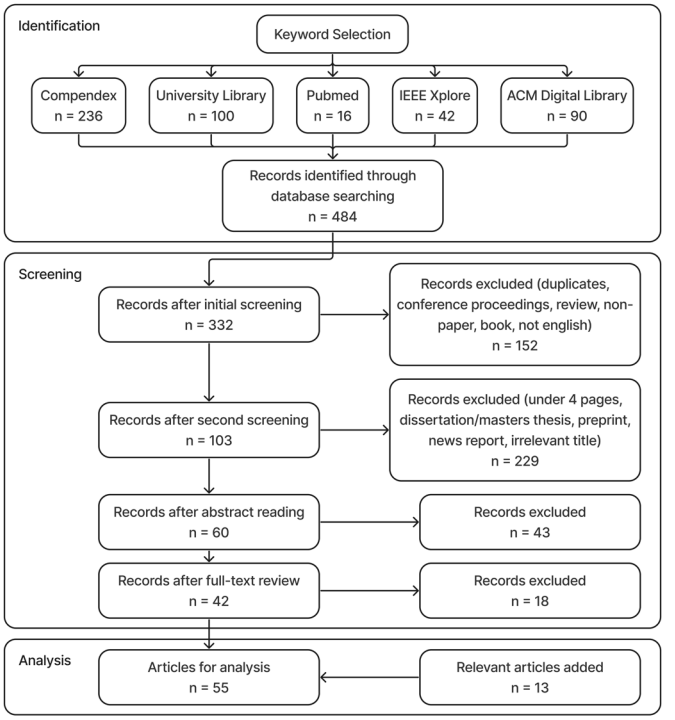

Fig. 1: The PRISMA diagram depicting the screening process.

The screening process began on February 4, 2025, and followed several phases. In the first screening, 152 records were excluded due to duplication or failure to meet basic inclusion criteria (e.g., not in English, published before 2015, not full papers, or published as short conference proceedings, reviews, or book chapters), resulting in 332 records. The second screening removed an additional 229 records that were preprints, theses, news reports, had irrelevant titles, and those under four pages, as these often lack sufficient methodological detail and results for robust synthesis, reducing the pool to 103 papers. Next, abstracts were screened for relevance, with 43 papers excluded for not including all three key components: an immersive technology (e.g., VR), a well-being or health-related focus, and a nature-based element. This brought the number of papers down to 60.

A detailed full-text review and classification process began on February 7, 2025, and was carried out over the next several weeks. During this phase, 18 papers were excluded for not using a head-mounted display (HMD), lacking a user study, or measuring unrelated outcomes. This resulted in a final set of 42 papers that met all criteria. An additional 13 papers were identified through exploratory searches and citation chaining, bringing the total number of studies included in this review to 55. All screening and inclusion/exclusion decisions were made by the first author to ensure consistent application of the criteria, which were discussed and agreed upon with the co-authors in advance. To further strengthen clarity and reliability, the co-authors reviewed the screening outcomes and contributed to refining the thematic classifications.

A summary of the selection process is depicted in the PRISMA flow diagram (Figure 1).

4 Classification Categories

To facilitate a high-level analysis of the 55 selected studies, we developed a structured classification framework that captures the diversity of research topics, technologies, outcomes, and methods in immersive virtual nature. Each category was selected to reflect key analytical dimensions for comparing user experiences and impacts.

- **Immersive Technology**: Captures the primary technological modality—VR or AR—used to deliver virtual nature experiences. Differentiating modalities helps assess fidelity, presence, and design constraints across platforms.

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- ***VR***: Fully simulated environments viewed through head-mounted displays (HMDs), enabling immersive sensory experiences.
- ***AR***: Digital overlays placed on real-world environments, typically used for mobile or wearable delivery.

- **Sensory Output Modality**: Classifies the sensory channels engaged——visual, auditory, or olfactory——to enhance immersion.
  - ***Visual***: Core *visual* content of natural environments.
  - ***Audio***: *Auditory* elements used to simulate natural soundscapes.
  - ***Olfactory***: *Scents* (e.g., essential oils) used to deepen immersion.
- **Thematic Focus/Application**: Groups studies by their dominant focus, which emerged through recurring outcome measures. These six themes took shape organically during the classification process, as recurring patterns in study objectives and measured outcomes revealed natural groupings around core emotional, physiological, and cognitive implications. The resulting structure was comprehensive, with all included studies fitting meaningfully into at least one of the thematic categories
  - ***Emotional Well-Being***: Studies focusing on improved emotional states such as happiness, mood, relaxation, or positive affect, typically measured through self-report scales.
  - ***Stress Reduction***: Studies targeting reductions in psychological or physiological stress markers (e.g., anxiety, heart rate, cortisol), often in response to acute stressors or using validated biometrics and questionnaires.
  - ***Cognitive Performance***: Studies measuring functions such as memory, attention, executive functioning, and processing speed through tasks or validated instruments.
  - ***Attention Recovery***: Studies assessing the restoration of attentional capacity after mental fatigue or overload, typically measured through concentration or focus-based tasks.
  - ***Restorative Benefits***: Studies investigating a general sense of psychological renewal, reduced mental fatigue, or increased vitality following immersive nature exposure.
  - ***Nature Connectedness***: Studies evaluating the emotional bond individuals feel toward the natural world, often measured through validated connectedness or relatedness scales.
- **Study Size (# of Participants)**: Documents participant count per study. Larger sample sizes offer greater generalizability, while variation may affect the interpretation of results.
- **Participant Profile**: Identifies specific populations (e.g., students, clinical patients) when reported. Understanding user vulnerability or baseline characteristics can help contextualize the outcomes.
- **Types of Measures**: Categorizes outcome data into three primary measures.
  - ***Psychological***: Self-report instruments for mood, anxiety, or well-being.
  - ***Physiological***: Biometrics (e.g., heart rate, EEG, cortisol) indicating bodily response.
  - ***Cognitive***: Objective metrics for memory, attention, reaction time, or executive function.
- **User Interaction**: Records whether users engaged interactively with the environment (e.g., navigation, virtual touch).
- **Exposure Duration**: Indicates how long participants were exposed to virtual nature environments.
  - ***Short-Term Exposure***: One-time or multi-session exposure over $\leq$2 days.
  - ***Long-Term Exposure***: Multi-week exposure with repeated sessions and follow-ups.

These classification categories guide our subsequent synthesis of how immersive virtual nature affects psychological states, cognitive functioning, and users' sense of connectedness to nature.

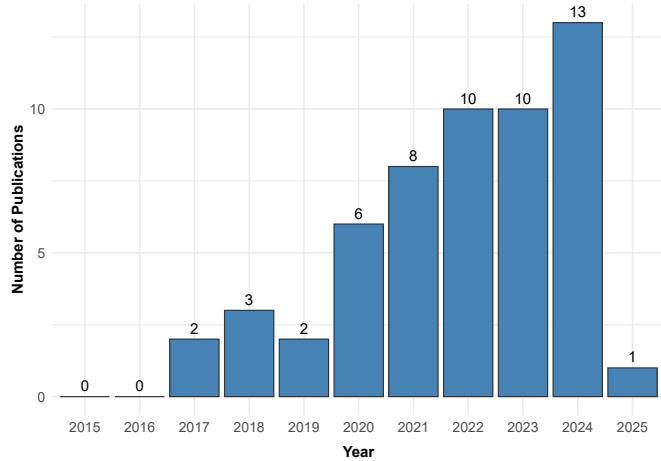

Fig. 2: Number of publications from 2015 to 2025.

5 Summary of Trends and Patterns

The classification process in this review provides a structured analysis of how VR/AR technologies are applied in virtual nature research to enhance well-being and nature connectedness. This approach highlights key trends and gaps in the field, offering insights into common research practices and areas needing further exploration. Notably, interest in this research area has grown substantially in recent years, as shown in Figure 2—the number of publications has steadily increased since 2015. The figures presented in this section provide a quantitative summary of these findings, contributing to a clearer understanding of the current research landscape and informing future studies on the effectiveness of virtual nature experiences. It is important to note that the classification categories are not necessarily mutually exclusive—for example, studies may address multiple sensory modalities and thematic areas. This overlapping classification reflects the multifaceted nature of virtual nature research, where a single intervention often targets several aspects of well-being simultaneously.

5.1 Immersive Technology

Among the 55 reviewed papers, nearly all utilized VR as the primary immersive technology, with only one paper incorporating AR alongside VR. This strong reliance on VR highlights its established role in delivering virtual nature experiences. In the singular instance involving AR, Tural & Tural [77] offered AR glasses as a sensory-friendly option for neurodiverse users, providing a less immersive experience via a virtual screen on an iPad. However, only 1 out of 35 participants opted to use this option. Although AR has seen limited use in our review, its potential along with that of other mixed reality (MR) technologies remains underexplored, pointing to a promising avenue for future research and contextually integrated design.

5.2 Sensory Output Modality

Different sensory output modalities can be involved in VR/AR simulations. In our review, all 55 papers employed visual output, underscoring the centrality of visual immersion in virtual nature experiences (Figure 3-left). Of these, 30 papers also incorporated auditory elements, creating more richly layered environments. In contrast, only two papers extended immersion to include olfactory output, and none systematically explored haptics. These findings illustrate a strong reliance on visual and audio modalities, with multimodal or sensory-integrated approaches beyond the visual-audio modalities still rare (Figure 3-right). The limited use of smell and other senses points to opportunities for enhancing realism and engagement by diversifying sensory inputs.

5.3 Thematic Focus/Application

The papers clustered around six key thematic applications, with many addressing multiple outcomes. Emotional Well-being (25 papers) and Stress Reduction (21 papers) were the most frequently investigated, suggesting a strong emphasis on affective and physiological health

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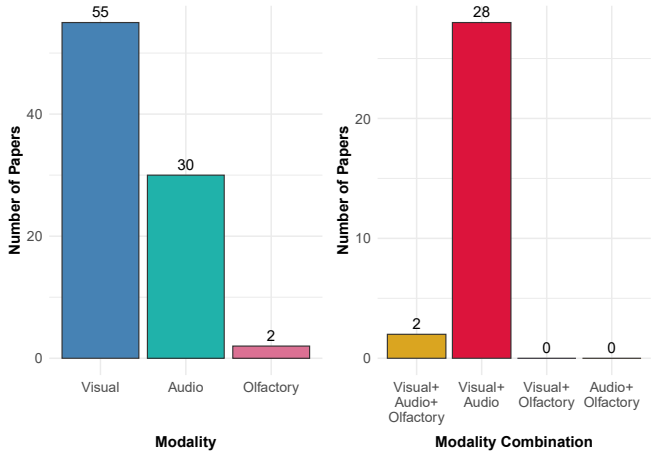

Fig. 3: Classification of sensory output modalities used in the reviewed studies. (Left) Frequency of papers employing visual, audio, and olfactory modalities individually. (Right) Number of papers using various combinations of these modalities.

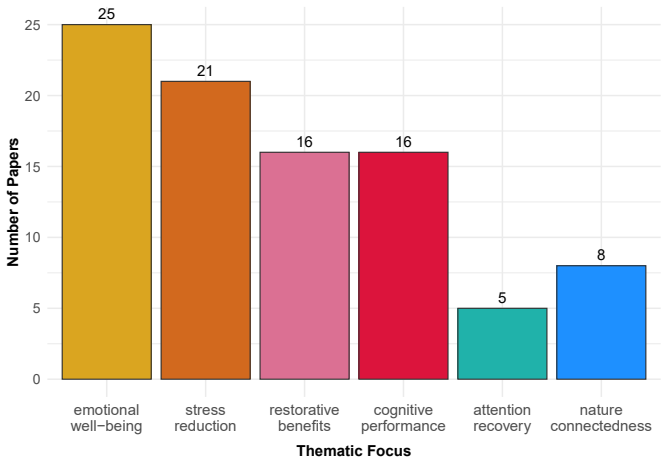

Fig. 4: Thematic focus of reviewed papers.

outcomes. Restorative Benefits (16 papers) and Cognitive Performance (16 papers) also featured prominently, showing the broader relevance of immersive nature for mental recovery and cognitive function. Fewer studies focused on Nature Connectedness (8 papers) and Attention Recovery (5 papers), though these areas are conceptually important and may represent emerging trends. Overall, the diversity of themes highlights virtual nature's multifaceted potential, while also revealing underrepresented domains ripe for further study. A detailed breakdown of these themes is presented in Figure 4.

5.4 Types of Measures

Our review examined which type of outcome measure was most commonly used in the literature across the three categories we defined: psychological, physiological, and cognitive. The results revealed that psychological measures were the most commonly used, appearing in 43 papers and capturing outcomes such as mood, stress, and perceived restoration. Physiological measures, including heart rate and skin conductance, were used in 29 papers, while cognitive measures such as attention or memory appeared in 18 studies. Many studies used combinations of these, enabling more comprehensive assessments of well-being and user impact. However, the lower frequency of physiological and cognitive measures indicates a methodological gap that future research could address by incorporating more objective and diverse assessments (see more details in Figure 5). An overview of which measures were used in each study is provided in the Supplementary Materials.

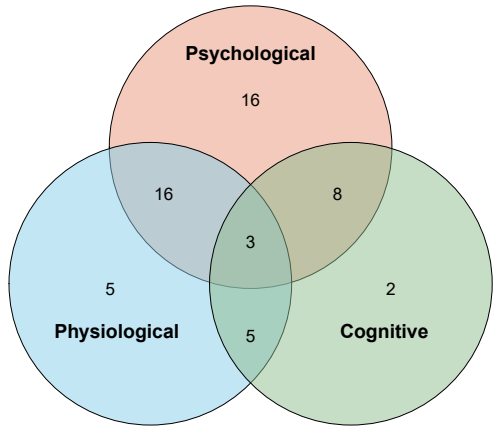

Fig. 5: Types and combinations of outcome measures used in the reviewed studies.

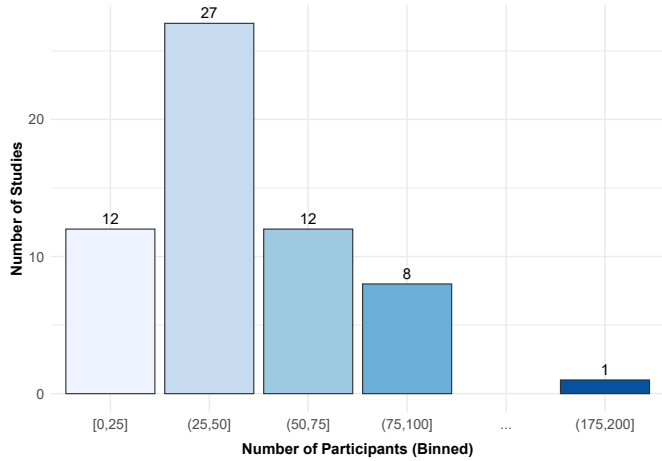

Fig. 6: Distribution of study sample sizes across the 60 user studies in 55 papers.

5.5 Study Size and Participant Profile

Across the 55 reviewed papers, a total of 60 user studies were conducted (as five papers included two studies each). Study sizes varied substantially, with participant numbers ranging from as few as 6 to as many as 198. The average number of participants was approximately 46 ($M = 46.28$, $SD = 31.28$). One study incorprated a large number of participants ($N = 198$), but the most common study size range was between 26 and 50 participants (27 of the 60 studies). This distribution suggests a tendency toward small to moderately sized samples, which is relatively typical for exploratory research involving immersive technologies and well-being outcomes (see the detailed distribution in Figure 6).

Regarding participant profiles, only a few papers specified their participant profiles—15 studies clearly identified targeted user groups (Figure 7). University students were the most frequently represented (7 studies), although many additional studies used them as a convenience sample without explicitly stating them as the focus. Clinical populations (e.g., medical patients) and older adults were each featured in 4 studies, highlighting some attention to vulnerable or high-need groups. However, children, working professionals, and individuals in low-access or rural environments remain largely underrepresented, pointing to a need for more inclusive and ecologically valid study designs.

5.6 User Interaction and Exposure Duration

In the context of this review, we defined user interaction as participants' ability to actively engage with or manipulate elements within the virtual environment beyond passive observation or simple navigation. Across the 55 reviewed papers, user interaction was minimal. Only two

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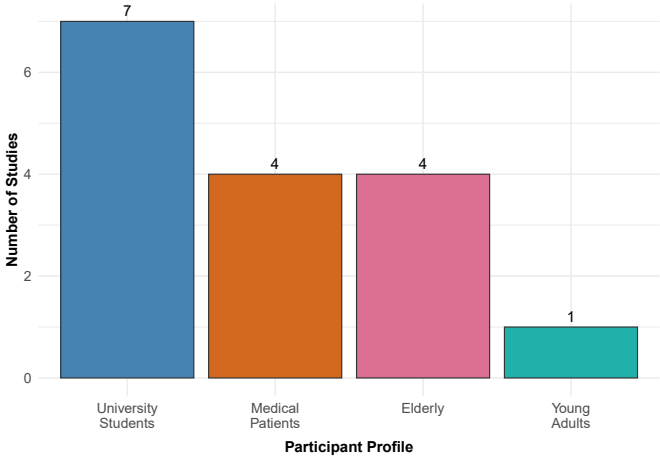

Fig. 7: Participant profiles reported in the reviewed papers. Note that not all papers reported detailed participant information.

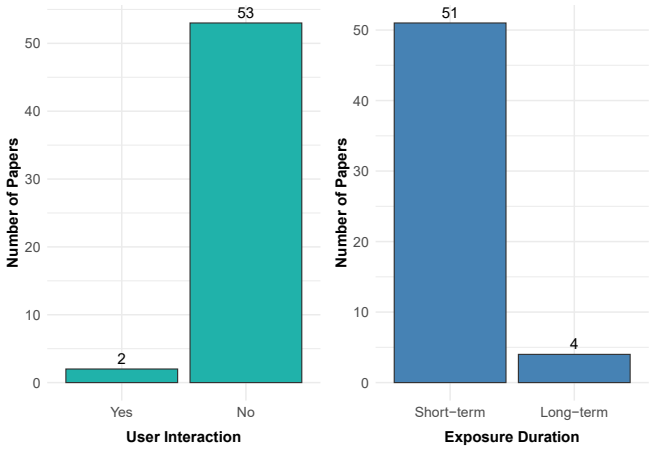

Fig. 8: User interaction and exposure duration across papers. (Left) Number of papers incorporating interactive versus passive experiences. (Right) Comparison of papers using short-term versus long-term exposure protocols, showing the dominance of short-term, non-interactive experiences.

papers incorporated interactive elements beyond basic locomotion or gaze-based navigation (Figure 8-left). The majority of papers adopted a passive exposure approach, where participants remained stationary and viewed pre-rendered or guided nature scenes without any direct influence on the environment. While a few papers allowed limited exploration (e.g., head movement in 360° video), these did not constitute meaningful interaction in the sense of responsive or embodied control. This limited interactivity may reflect both methodological constraints and a prioritization of observational immersion over active engagement, which should be discussed further.

Exposure duration was similarly constrained in most papers. Of the 55 papers, 51 implemented short-term exposure protocols, typically involving single sessions lasting between 5 and 30 minutes (Figure 8-right). Only four papers featured long-term exposure, defined here as multi-session designs spanning several days or weeks. While short exposures were sufficient to detect acute effects on stress, mood, or perceived restoration, they offer limited insight into the sustainability of these benefits or the impact of repeated virtual nature engagement.

6 In-Depth Thematic Reviews

This section presents a comprehensive qualitative review of the collected papers, focusing on the thematic categories introduced in Section 4: *Emotional Well-Being*, *Stress Reduction*, *Restorative Benefits*, *Cognitive Performance*, *Attention Recovery*, and *Nature Connectedness*. The first three themes are interrelated in the context of well-being and health, while the next two pertain to perceptual and cognitive performance. The final theme is conceptually linked to the others through its emphasis on emotional attachment to and interest in nature. We review key findings and measurement approaches from selected studies, with a focus on their impacts and research contexts. While some conceptual overlap exists among the themes, each paper was allocated to a single category that best represented its primary focus. This approach supported thematic clarity and helped ensure a balanced distribution of studies across all six categories. A complete list of all reviewed papers and their thematic classifications is provided in the Supplementary Materials.

6.1 Emotional Well-being

Emotional well-being, defined as the experience of positive emotional states, emerged as the most common outcome across the studies reviewed, appearing in 25 papers. This prevalence is not surprising given that immersive technologies are frequently used for enjoyment and entertainment, which naturally prioritize affective experience over clinical outcomes. While emotional well-being is undoubtedly part of overall health, it is often approached in these studies through the lens of user experience and engagement rather than through standardized medical or physiological measures. That said, many studies in this category point to meaningful benefits for relaxation, mood regulation, and psychological comfort.

To assess these emotional benefits, researchers employed both psychological and physiological indicators. Psychological measures formed the backbone of this research area. Tools like the Positive and Negative Affect Schedule (PANAS) [80], which captures short-term mood states across two dimensions: positive and negative affect, the State-Trait Anxiety Inventory (STAI) [75], which distinguishes between momentary (state) and more enduring (trait) anxiety levels, the Zuckerman Inventory of Personal Reactions (ZIPERS) [95], which evaluates affective and sensory responses to environmental stimuli, and qualitative interviews were used to assess shifts in mood, anxiety, and overall affect. Chin et al. [12] found that women with metastatic breast cancer experienced significant improvements in fatigue and depressive symptoms after daily exposure to nature-based VR. Navarrete et al. [58] and Lundstedt et al. [44] similarly observed improved emotional states among confined or older adult users, with participants describing the VR experiences as calming, enjoyable, or emotionally uplifting. Mahrous et al. [47] and Li et al. [41] found reduced stress and negative emotion in cognitively demanding, confined VR settings, highlighting the regulatory potential of immersive nature design. In Firth & Jayadas [20], participants in a spaceflight simulation reported enhanced emotional comfort and connection following exposure to biophilic VR designs.

To support participants' self-reported emotional responses, several studies also used physiological metrics. Jung et al. [33] used EEG to detect decreased emotional arousal and increased frontal alpha activity when participants were exposed to biophilic VR environments, indicating a calm, relaxed state. In addition, Hasa & Husein [3] incorporated physiological metrics such as heart rate and blood pressure, suggesting a link between immersive biophilic design and well-being. Although cognitive performance was not the primary measure in this category, a few studies suggested that emotional regulation in VR nature contexts may indirectly support clarity or attentional resilience. For example, Mostajeran et al. [56] reported that participants exposed to immersive forest scenes experienced enhanced positive affect and demonstrated improved attention, concentration, and working memory, suggesting cognitive benefits.

Taken together, these findings reinforce the role of virtual nature as a valuable tool for enhancing emotional well-being across diverse populations and contexts. The integration of self-report measures and physiological data adds further validity to these outcomes. At the same time, several studies point to important limitations and areas for further inquiry. Verzwyvelt et al. [79], for instance, found no significant physiological effects, highlighting variability in emotional responses and the need for more sensitive or standardized measures. Similarly, Emamjomeh et al. [17] reported no significant outcomes of biophilic over non-biophilic VR designs, raising questions about whether immer-

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sive technology offers unique advantages. These findings suggest that while early evidence is promising, future research should clarify the distinct contributions of VR nature and refine the design elements that most effectively support emotional well-being.

6.2 Stress Reduction

Stress reduction was the second most frequently reported outcome across the reviewed studies with 21 papers, highlighting its emerging relevance in healthcare-related applications of VR (and AR). As the field increasingly explores VR as a tool with therapeutic potential, stress mitigation offers a compelling use case, particularly in clinical environments where physiological stress responses are closely linked to patient outcomes. While emotional well-being reflects general mood enhancement and engagement, stress reduction targets biological systems tied to health risks and carry long-term health implications. To evaluate the impact of VR nature experiences on stress, studies employed a combination of physiological, psychological, and cognitive indicators. These measures captured both objective bodily responses and subjective perceptions of stress relief across clinical and non-clinical contexts.

Physiological measures offered the most direct assessment of stress responses, particularly those associated with sympathetic nervous system activity. Common indicators included heart rate (HR), blood pressure (BP), heart rate variability (HRV), electrodermal activity (EDA), and EEG signals. Multiple studies reported significant reductions in physiological arousal during or after exposure to immersive VR that simulated natural settings. Latini et al. [37] showed that audio-visual congruence—such as natural sounds paired with indoor greenery—led to significant drops in EDA and pulse rate, particularly in office settings. In a comparison of VR and biofeedback, Kim et al. [35] found that VR produced greater parasympathetic activation, evidenced by favorable shifts in HRV and LF/HF ratios. Jo et al. [32] showed that forest-based VR improved HRV, while Ren et al. [65] demonstrated that even short aquarium-based VR exposures could significantly reduce HR in university students. Yin et al. [88] reported reductions in BP during exposure to VR nature, while Yeom et al. [87] found that green wall environments in VR decreased stress-related EEG activity.

Psychological and cognitive measures captured how participants perceived stress, emotional relief, and cognitive clarity. These were assessed using tools such as the Perceived Stress Scale (PSS) [14], which measures the degree to which situations in one's life are appraised as stressful, State-Trait Anxiety Inventory (STAI) [75], and visual analog scales (VAS) [82], which allow participants to rate their current emotional or mental states along a continuum, as well as self-report interviews. For example, Zhang et al. [94] found that VR environments with immersive greenery improved self-perceived stress and supported cognitive recovery. Xiaoxue & Huang [85] similarly observed stress reduction and anxiety recovery among participants exposed to simulated green walls and outdoor views. Yin et al. [89] linked lower perceived stress to higher mental restoration in biophilic VR office settings, pointing to the potential for immersive environments to enhance attentional readiness. Together, these findings emphasize the subjective and cognitive dimensions of stress relief, which may operate independently of biometrics but are no less meaningful in health or workplace settings.

Despite encouraging results, several studies pointed to meaningful limitations that should guide future research. Abbott & Diaz-Artiles [1] explored scent-enhanced VR nature scenes and reported promising reductions in anxiety, but the small sample size (N=10) limits generalizability. Backtash et al. [4] found only marginal advantages of immersive VR over traditional imagery, raising questions about the added value of sensory-rich design. Scates et al. [69] highlighted a common discrepancy between subjective relaxation and modest reductions in stress and pain scores, suggesting the need for more integrated evaluation frameworks. These examples point to the importance of aligning outcome measures, diversifying participant samples, and refining intervention design.

6.3 Restorative Benefits

Restorative benefits were identified in 16 studies, making this a well-represented and conceptually rich theme across the reviewed literature. While this category overlaps with emotional well-being, stress reduction, and attention recovery, it reflects a distinct type of outcome commonly reported in immersive virtual nature research. Emotional well-being emphasizes affective states, stress reduction targets physiological arousal, and attention recovery focuses on the replenishment of directed attention. Restorative benefits, by contrast, describe a broader sense of psychological relief—often characterized by mental clarity, reduced cognitive load, or a feeling of being "refreshed." Although both restorative benefits and attention recovery are grounded in Attention Restoration Theory (ART) [34], the former typically captures subjective impressions of recovery, while the latter is measured more directly through cognitive tasks or attentional performance. Most studies in this category draw from ART, which posits that exposure to nature helps replenish depleted attentional resources, or Stress Recovery Theory (SRT) [78], which suggests that natural environments promote a return to physiological and psychological baseline after stress. Notably, many papers explicitly identify "restoration" or "restorativeness" as a primary goal, reinforcing its relevance as a standalone implication.

Restorative outcomes were most commonly measured using subjective tools such as the Perceived Restorativeness Scale (PRS) [29], a self-report measure designed to evaluate how effectively an environment supports mental restoration, along with items assessing clarity, coherence, and the sense of psychological escape. Nicoly et al. [59] examined the influence of visual motion and environmental design, showing that naturalistic movement enhanced perceived restorativeness. Several studies emphasized spatial and architectural features; for example, Tural & Tural [77] demonstrated that biophilic features, such as window views, consistently enhanced restorative quality in architectural simulations. Lyu et al. [45] demonstrated that thermal comfort and biophilic design elements in semi-outdoor workplace environments significantly influenced how restorative participants perceived the space to be, while Mattila et al. [50] and Yu et al. [93] showed that virtual forest environments supported perceived restoration and improved mood compared to less natural alternatives. Masters et al. [49] further found that higher realism in virtual forest scenes significantly enhanced perceived restorativeness. Together, these findings suggest that immersive VR environments can effectively elicit perceived restoration, particularly when biophilic design, environmental realism, and multisensory cues are carefully considered.

Although restorative benefits are most often evaluated through subjective reports, fewer studies include physiological and cognitive indicators to support these findings. Yoon & Jeon [91] reported that immersive audio-visual nature scenes improved both self-reported and physiological measures of restoration. Lyu et al. [46] extended this approach in an experimental setting, observing improved cognitive performance and perceived restoration following exposure to a virtual semi-outdoor roof garden enhanced with sunlight, shade, and greenery. These studies suggest restoration is a multifaceted experience blending felt relief with physiological change. Continued research would benefit from incorporating performance-based cognitive tasks to clarify links between perceived restoration and cognitive functioning.

Restorative benefits represent a unique but integrative outcome in immersive virtual nature research. While they overlap conceptually with emotional, cognitive, and physiological outcomes, their emphasis on subjective mental clarity and psychological recovery distinguishes them as a meaningful standalone measure. Many studies have leveraged this framework to capture experiential outcomes that are not always reflected in performance or clinical indicators. However, inconsistencies remain in how restorativeness is measured, and the effectiveness of VR nature appears to depend heavily on design quality and context. Masters et al. [48] observed trends suggesting that high-biomass nature environments may be more restorative than low-biomass ones, though the difference was not statistically significant. Findings from Reese et al. [64] and Browning et al. [8] indicate that virtual nature can enhance perceived restorativeness and affect compared to non-nature controls, but sometimes underperforms relative to real nature, even during short exposures. Future research should explore how virtual environments support psychological renewal over time and which immersive features sustain it.

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6.4 Cognitive Performance

Cognitive performance refers to measurable improvements in functions, such as memory, attention, executive processing, and creativity following exposure to virtual nature. Although relatively fewer studies in this review focused explicitly on cognitive outcomes (16 papers), compared to those on emotional well-being and stress reduction, this application holds strong potential, particularly in education, healthcare, and performance-driven environments. As immersive technologies become more accessible, VR nature offers a promising tool for supporting mental clarity, task engagement, and productivity in settings where cognitive performance is critical. This is especially relevant in an increasingly urbanized world and within hybrid or digital workspaces, where natural stimuli are often limited. By offering the cognitive benefits of nature without requiring physical access, VR may help bridge this environmental gap and enhance focus and learning in classrooms, workplaces, and rehabilitation settings.

Several studies in this theme used psychological and physiological indicators to infer cognitive engagement and mental workload during or after VR nature exposure. Winn et al. [84] measured participants' self-reported mental workload using the NASA-TLX [28] and found that immersive VR nature scenes in a space habitat simulation reduced perceived cognitive load during task execution. Sharam et al. [71] investigated how virtual windows with natural views influence cognitive performance and affect, finding that exposure to natural environments in VR improved participants' creative fluency and positive mood during performance tasks. Gattulo et al. [23] complemented self-report measures with eye-tracking data, showing that users in biophilic VR training simulations exhibited more focused visual attention patterns, suggesting greater engagement and reduced cognitive distraction.

More direct effects were observed in studies that used performance-based cognitive tasks. Palanica et al. [63] applied the Alternate Uses Test (AUT) [27] to evaluate creativity and found that both 2D and VR nature conditions enhanced divergent thinking, with improvements in fluency, flexibility, and originality. Yin et al. [90] employed the visual backwards digit span test and observed improvements in short-term memory capacity after participants engaged with biophilic environments. Martínez Manchón & Šimunić [15] further demonstrated that users navigating biophilic VR classroom simulations improved mood and reduced stress, potentially supporting a more favorable mental state for cognitive engagement.

However, not all findings were uniformly positive. Mostajeran et al. [57] used a memory recall task embedded in a redirected walking exercise and found no significant performance differences between forest and urban environments, suggesting that environmental content alone may not influence all cognitive outcomes. Romano et al. [66] incorporated self-reported distraction levels and working memory tests to evaluate animated biophilic features in VR but found that while these increased immersion, they also raised perceived cognitive load and did not improve memory performance. These mixed results suggest that the cognitive effects of VR nature may depend on task complexity, sensory design, and user expectations. While the studies reviewed suggest that virtual nature holds meaningful potential for enhancing cognitive performance, this application remains underexplored in the context of VR for health outcomes. As immersive nature integrates into wellness and education, targeted research is needed to assess its impact on cognitive load, memory, and executive function in applied contexts.

6.5 Attention Recovery

Attention recovery refers to the restoration of directed attention capacity following cognitive fatigue. While conceptually related to cognitive performance, it stands apart as a distinct benefit observed in a smaller subset of studies (5 papers). Grounded in ART [34], attention recovery involves regaining the ability to focus, sustain concentration, and filter out distractions after prolonged mental effort. These functions are increasingly relevant in a world dominated by short-form media, multitasking, and constant digital stimulation—factors that have contributed to declining attention spans and difficulty maintaining focus. Though not a benefit users typically associate with VR, these studies suggest that immersive virtual nature experiences may offer an effective way to restore attention.

Across the studies reviewed, attention recovery was evaluated using a mix of physiological, psychological, and cognitive indicators. Physiological measures were used in two studies to assess changes in bodily states linked to attentional readiness. Gerber et al. [25] measured heart rate, mean arterial pressure, and eye-tracking, finding that VR nature scenes reduced physiological arousal and promoted sustained visual engagement. Similarly, Anderson et al. [2] monitored electrodermal activity (EDA) and affective states, reporting decreased arousal following VR exposure in a simulated confinement environment, suggesting a return to a more attention-supportive state. Psychological measures focused on subjective perceptions of mental clarity and attentional fatigue. Chan & Huang [10] measured attention recovery using subjective ratings after participants experienced immersive biophilic VR, finding that the environments supported attention restoration. Li et al. [40] also used self-report measures during a mobile VR commute, where participants noted slight improvements in focus and emotional state when exposed to natural scenes. Cognitive measures were used by Fang et al. [18], who employed the Sustained Attention to Response Task (SART) to directly assess attentional performance. Their results showed that VR breakrooms with a moderate plant density significantly improved attention task scores, suggesting an optimal level of greenery for enhancing cognitive functioning. Together, these findings point to early but promising evidence that virtual nature may play a role in helping users restore attentional control after cognitive fatigue.

Although attention recovery was addressed in only a handful of studies, the findings highlight a promising and underexplored application of virtual nature. This work suggests that VR may help counteract the cognitive fatigue and fragmented focus associated with modern digital environments. While early evidence shows that VR nature experiences can enhance attentional readiness across physiological, psychological, and behavioral domains, this area remains largely overlooked in VR and health research. Future studies should explore how virtual nature can reliably support attentional rebound and sustained focus, particularly as distractions become an increasingly dominant feature of daily life. These early findings highlight VR nature's potential to alleviate attentional fatigue, particularly in digitally distracting environments.

6.6 Nature Connectedness

Nature connectedness holds a distinct position within immersive virtual nature research. While many papers reference the importance of reconnecting humans with nature, only a small subset directly assessed this construct (8 papers). Unlike implications such as emotional well-being or stress reduction, which reflect temporary states, nature connectedness involves a more stable sense of self in relation to the natural world. It captures how individuals identify with, feel part of, and relate to nature on a psychological level, often reflecting deeper shifts in perspective or values. Although conceptually adjacent to other outcomes, it is treated here as a standalone implication due to its emphasis on enduring identity rather than transient affect. Its inclusion in VR nature research signals an important shift toward understanding not just how people feel during immersive experiences, but how those experiences may reshape their orientation to the natural world in more lasting ways.

Among the studies that explicitly measured nature connectedness, outcomes were primarily assessed using psychological self-report tools, with limited inclusion of physiological or cognitive data. Most studies used validated scales such as the Connectedness to Nature Scale (CNS), Nature Relatedness Scale (NR), or Inclusion of Nature in Self (INS) diagram to capture shifts in participants' perceived relationship with the natural world [51,60,70]. Leung et al. [39] and Finkler et al. [19] found that even brief immersive VR sessions could enhance self-reported connectedness, regardless of participants' baseline nature affinity. In educational contexts, You et al. [92] and Chan et al. [11] evaluated immersive biophilic environments in educational settings and observed post-exposure increases in CNS and INS scores, suggesting that VR classrooms and forest environments can help reinforce nature identity in learners. Sansal et al. [68] extended this to elder care contexts, showing that brief VR nature exposure significantly improved emotional well-

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being in older adults with limited real-world access to greenspace, despite having limited real-world access to greenspace. Additionally, Yeo et al. [86] compared delivery formats, revealing that interactive or embodied experiences were more effective than passive experiences in enhancing nature connectedness, and Smith et al. [73] compared real and virtual nature experiences, finding that while both formats supported nature connectedness, the virtual condition led to greater increases in experiential connection for some participants. McKeever et al. [52] used a VR version of the "Overview Effect" to evoke feelings of awe and existential perspective shift, suggesting that immersive design may foster deeper psychological shifts beyond traditional scale-based measures.

While nature connectedness research is predominantly psychological, a few studies explored physiological or cognitive dimensions. Chan et al. [11] used ECG monitoring and found that reductions in physiological arousal occurred alongside increases in self-reported nature connectedness. You et al. [92] explored both cognitive and physiological changes using fNIRS and behavioral measures, but found no significant changes—highlighting the need for more targeted measures or refined experimental designs. Notably, none of the studies included structured cognitive performance tasks (e.g., working memory, sustained attention), leaving open how identity-level nature affiliation interacts with attentional or cognitive processing over time.

While fewer studies have measured nature connectedness directly, those that have reveal its potential as a meaningful and underutilized outcome in immersive VR research. Unlike affective or performance-based measures, connectedness speaks to more enduring shifts in identity, empathy, and environmental affiliation. The consistent use of validated self-report tools demonstrates growing interest in this area, but the limited integration of physiological data and absence of cognitive performance tasks present opportunities for further exploration. As digital environments increasingly mediate human-nature relationships, future research should explore how immersive experiences might cultivate deeper, lasting forms of connection—particularly in populations with limited access to real-world nature. Incorporating longitudinal designs and diverse cultural perspectives will also be essential to understanding how virtual nature can influence not only how users feel, but who they believe themselves to be in relation to the natural world.

7 Discussion

Based on the high-level classification results and the detailed reviews described above, this section summarizes our key findings related to the research questions introduced in Section 1.

7.1 Summary of Findings and Implications

7.1.1 Use of Immersive Technologies (RQ1)

Dominance of VR and Limited Use of AR: Nearly all studies reviewed employed VR, with only one integrating AR. While VR's immersive capabilities and maturity make it a dominant platform, AR (and MR) remains underutilized despite offering critical affordances for real-world, situated nature experiences. These include lightweight delivery, context-aware augmentation, and integration into daily environments like classrooms or hospital rooms. Emerging research should explore how AR/MR can deliver contextually embedded, biophilic interventions. Studies should examine in-situ applications (e.g., nature overlays in homes/offices) and how spatial grounding affects restoration and connectedness, particularly in constrained or urban settings.

Sensory Modality Gaps: While all studies included visual output and most included audio, only two studies incorporated olfactory cues, and none used haptics. This overreliance on visual-audio channels limits the fidelity and affective depth of virtual nature experiences, especially given the rich multisensory nature of real-world environments. Expanding sensory design is a key research priority. Subsequent work should investigate how multisensory cues influence immersion, emotional regulation, and memory, as they can enrich user experience and presence but are often underutilized due to practical constraints [53].

Limited Interactivity: Only two studies included meaningful interactivity beyond passive viewing or gaze-based navigation. This stands in contrast to broader VR/AR and HCI research, where interaction is essential to agency, learning, and personalization. The lack of embodied, goal-directed, or exploratory interaction likely reduces engagement and may limit the depth of affective and cognitive benefits. Future studies should incorporate embodied, interactive experiences—such as walking, object manipulation, or natural gesture input interacting with virtual nature contents. Research should examine how such interactivities foster agency, engagement, and personalization, and how these affordances shape emotional regulation, cognitive load, and nature connectedness.

7.1.2 Thematic Trends and Outcomes (RQ2)

Thematic Prioritization on Emotional Well-Being and Stress Reduction: Emotional well-being and stress reduction were the most frequently covered themes, reflecting the field's early focus on short-term affective benefits. In contrast, cognitive performance and especially attention recovery were relatively underexplored. This may be due to the greater ease of assessing emotional states through validated self-report tools, whereas cognitive performance requires more complex, task-based methods. Early immersive nature studies also tended to draw from therapeutic rather than cognitive or performance-based frameworks. This gap presents an opportunity. Immersive nature aligns closely with ART, which suggests that exposure to restorative environments can help replenish mental resources. In an era of widespread digital fatigue, immersive nature may offer a valuable tool for supporting attention and executive function. Future research should prioritize these outcomes, using immersive-compatible cognitive tasks, objective performance metrics, and real-world testing in educational, workplace, or rehabilitation contexts.

Nature Connectedness as a Latent Mechanism: As discussed in Section 6.6, nature connectedness emerged not only as a distinct outcome but also as a latent mechanism linked to emotional and cognitive benefits. Although only a small subset of studies measured it directly, several suggestive patterns—particularly among users with low baseline affinity to nature—point to its mediating potential. Chovanec et al. [13] also highlights this gap, noting that the absence of specific measurements on immersive technology's ability to enhance nature connectedness represents a critical area for further empirical exploration, particularly given the current lack of quantification in this domain. Future research should build on these observations by incorporating validated connectedness measures and mediation-based models, especially in longitudinal designs that assess sustained psychological change.

7.1.3 Methodological Limitations and Research Gaps (RQ3)

Underrepresentation of Diverse Populations and Real-World Contexts: Across the reviewed studies, participants were predominantly university students, with most research conducted in controlled lab. This narrow demographic and contextual scope raises concerns about generalizability and equity. Populations who may benefit most from virtual nature—such as older adults, children, neurodivergent individuals, or those in low-resource environments—remain largely underrepresented. Furthermore, the limited ecological validity of lab-based studies restricts understanding of how immersive nature functions in everyday scenarios, where environmental variability, interruptions, and device constraints are common. Future studies should use carefully controlled designs that compare virtual nature with other VR conditions to isolate its specific effects from the general impact of VR immersion. In addition, research should broaden participant diversity and test immersive nature across a wider range of real-world contexts—including classrooms, workplaces, homes, and public environments. Emphasizing ecologically valid study designs will improve the translational potential of immersive nature interventions and ensure their accessibility and effectiveness for underserved and high-need populations.

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Lack of Longitudinal Research: The majority of studies reviewed relied on brief, single-session interventions—typically lasting less than 30 minutes. While these designs are practical for assessing immediate effects on mood or stress, they provide limited insight into how immersive nature experiences function over time, integrate into daily routines, or contribute to lasting psychological change. Few studies examined repeated exposure, and even fewer assessed whether immersive nature experiences produce cumulative benefits, sustained engagement, or habituation effects. Future research should prioritize longitudinal designs that track immersive nature use over days, weeks, or months. Studies should assess how repeated exposure influences outcomes such as stress resilience, cognitive function, emotional regulation, and nature connectedness. Additionally, thematic benefits such as restoration and identity development may require prolonged engagement or interaction to fully manifest. Researchers should explore real-world integration—e.g., using daily micro-interventions like 5-minute VR breaks—and investigate whether benefits grow, stabilize, or diminish over time. Combining long-term tracking with follow-up assessments and qualitative feedback will yield more ecologically valid and behaviorally relevant insights.

7.2 Limitations

While this review provides a comprehensive synthesis of immersive virtual nature research, several limitations should be noted. First, although we employed systematic criteria for inclusion, our search was limited to peer-reviewed studies published in English. As a result, relevant work published in other languages or in grey literature (e.g., theses, technical reports) may have been omitted, potentially skewing the representativeness of the findings.

Second, the scope of the review was intentionally focused on immersive technologies using HMDs. Consequently, studies involving non-immersive or screen-based contents, while relevant for comparison, were excluded. This focus allowed for depth but may limit the generalizability to broader digital nature applications.

Third, while the thematic coding process was iterative and cross-validated among authors, it inherently involved interpretive judgment. Some degree of subjectivity in theme identification, categorization, and emphasis is unavoidable, especially in synthesizing diverse study designs, user populations, and outcome measures. Although an independent second screening would have strengthened the rigor of the process, the first author carried out the screening, with the criteria and results subsequently reviewed by the co-authors to maintain clarity and consistency. In addition, a formal risk of bias assessment was not performed given the descriptive scope and heterogeneity of studies, but detailed reporting is provided to help readers assess the strength of the evidence.

Finally, the rapidly evolving nature of immersive technology means that this review represents a snapshot of a fast-changing field. New hardware platforms, sensory integration methods, and application domains are emerging quickly and may not yet be reflected in the literature captured during our review period.

8 Conclusion

This paper provides an integrative review of 55 research papers investigating immersive virtual nature experiences. Across diverse domains—including emotional well-being, stress reduction, cognitive performance, and nature connectedness—the findings generally highlight the potential of immersive nature to support a range of psychological and physiological outcomes. However, the field remains unevenly developed, with most studies relying on VR, visual-auditory modalities, short-term exposures, and relatively homogeneous user populations. The review also revealed several system-level design limitations, such as limited interactivity and a lack of multisensory integration.

Addressing these gaps—particularly in population diversity, sensory design, cognitive modeling, and long-term engagement—will be essential for advancing the field. Future research can help shape immersive nature experiences that are not only more effective, but also inclusive, sustainable, and contextually relevant across domains of health, education, and beyond.

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References

[1] R. W. Abbott and A. Diaz-Artiles. The impact of digital scents on behavioral health in a restorative virtual reality environment. *Acta Astronautica*, 197:145–153, 2022. doi: 10.1016/j.actaastro.2022.05.025 6

[2] A. P. Anderson, M. D. Mayer, A. M. Fellows, D. R. Cowan, M. T. Hegel, and J. C. Buckey. Relaxation with immersive natural scenes presented using virtual reality. *Aerospace Medicine and Human Performance*, 88(6):520–526, June 2017. doi: 10.3357/AMHP.4747.2017 7

[3] A. Baiz and H. Husein. Promoting students' well-being indicators through adapting biophilic design attributes in salahaddin university dormitories. *Al-Qadisiyah Journal for Engineering Sciences*, 16:180–186, 10 2023. doi: 10.30772/qjes.2023.179985 5

[4] H. Baktash, D. Kim, and A. Shirazi. Beyond sight: Comparing traditional virtual reality and immersive multi-sensory environments in stress reduction of university students. *Frontiers in Virtual Reality*, 5, 2024. doi: 10.3389/frvir.2024.1412297 6

[5] M. G. Berman, J. Jonides, and S. Kaplan. The cognitive benefits of interacting with nature. *Psychological science*, 19(12):1207–1212, 2008. doi: 10.1111/j.1467-9280.2008.02225.x 1

[6] E. Brambilla, E. Petersen, K. Stendal, V. Sundling, T. E. MacIntyre, and G. Calogiuri. Effects of immersive virtual nature on nature connectedness: A systematic review protocol. *Digit. Health*, 8:20552076221120324, Jan. 2022. 2

[7] G. N. Bratman, G. C. Daily, B. J. Levy, and J. J. Gross. The benefits of nature experience: Improved affect and cognition. *Landscape and urban planning*, 138:41–50, 2015. doi: 10.1016/j.landurbplan.2015.02.005 1

[8] M. H. E. M. Browning, K. J. Mimnaugh, C. J. van Riper, H. K. Laurent, and S. M. LaValle. Can simulated nature support mental health? comparing short, single-doses of 360-degree nature videos in virtual reality with the outdoors. *Front. Psychol.*, 10:2667, 2019. 6

[9] C. A. Capaldi, R. L. Dopko, and J. M. Zelenski. The relationship between nature connectedness and happiness: a meta-analysis. *Frontiers in Psychology*, 5, 2014. doi: 10.3389/fpsyg.2014.00976 1

[10] H. Chan and Y. Huang. Research on the benefits of biophilia effects in virtual environments. In J. Y. C. Chen and G. Fragomeni, eds., *Virtual, Augmented and Mixed Reality*, pp. 16–28. Springer Nature Switzerland, Cham, 2024. 7

[11] S. H. M. Chan, L. Qiu, G. Esposito, K. P. Mai, K.-P. Tam, and J. Cui. Nature in virtual reality improves mood and reduces stress: evidence from young adults and senior citizens. *Virtual Real.*, pp. 1–16, Nov. 2021. 7, 8

[12] S. Chin, A. Cavadino, A. Akroyd, G. Tennant, R. Dobson, A. Gautier, and L. Reynolds. An investigation of virtual reality nature experiences in patients with metastatic breast cancer: Secondary analysis of a randomized controlled trial. *JMIR Cancer*, 8(3):e38300, July 2022. 5

[13] S. Chovanec, S. Gschanes, B. Tölgyesi, and A. K. Bakk. Augmented reality's role in strengthening human-nature connections: A systematic review. In *Proceedings of the 27th International Academic Mindtrek Conference*, Mindtrek '24, 8 pages, p. 286–293. Association for Computing Machinery, New York, NY, USA, 2024. doi: 10.1145/3681716.3689450 2, 8

[14] S. Cohen, T. Kamarck, R. Mermelstein, et al. Perceived stress scale. *Measuring stress: A guide for health and social scientists*, 10(2):1–2, 1994. 6

[15] F. de Asís Martínez Manchón and A. Šimunić. Effects of short-term exposure to interactive vs. non-interactive virtual nature on cognitive performance and mental health in college students. *International Journal of Human–Computer Interaction*, 40(18):5443–5453, 2024. doi: 10.1080/10447318.2023.2235122 7

[16] E. Di Giuseppe, L. Marcelli, A. Latini, and M. D'Orazio. Evaluating biophilic design strategies in immersive virtual indoor environments: A systematic review on the implications for buildings occupants. *Journal of Building Engineering*, 98:111127, 2024. doi: 10.1016/j.jobe.2024.111127 2

[17] A. Emamjomeh, Y. Zhu, and M. Beck. The potential of applying immersive virtual environment to biophilic building design: A pilot study. *Journal of Building Engineering*, 32:101481, 2020. doi: 10.1016/j.jobe.2020.101481 5

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[18] Z. Fang, H. Xu, L. Tao, Y. Tan, Y. Li, and J. Wu. Effects of volume ratio, layout and leave size of indoor plants on workers' attention recovery in factory staff break area. *Buildings*, 13(3):622, Feb. 2023. 7

[19] W. Finkler, L. Vlietstra, D. L. Waters, L. Zhu, S. Gallagher, R. Walker, R. Forlong, and Y. van Heezik. Virtual nature and well-being: Exploring the potential of 360° VR. *Appl. Psychol. Health Well Being*, 17(1):e70008, Feb. 2025. 1, 7

[20] A. Firth and A. Jayadas. Biophilic design of the iss crew quarters to improve cognitive and physiological health measures. In *2022 IEEE Aerospace Conference (AERO)*, pp. 1–10, 2022. doi: 10.1109/AERO53065.2022.9843279 5

[21] S. Frost, L. Kannis-Dymand, V. Schaffer, P. Millear, A. Allen, H. Stallman, and J. Atkinson-Nolte. Virtual immersion in nature and psychological well-being: A systematic literature review. *Journal of Environmental Psychology*, 80:101765, 2022. doi: 10.1016/j.jenvp.2022.101765 2

[22] J. S. Gaekwad, A. Sal Moslehian, and P. B. Roös. A meta-analysis of physiological stress responses to natural environments: Biophilia and stress recovery theory perspectives. *Journal of environmental psychology*, 90:102085–, 2023. 2

[23] M. Gattullo, E. Laviola, S. Romano, A. Evangelista, V. M. Manghisi, M. Fiorentino, and A. E. Uva. Biophilic enriched virtual environments for industrial training: a user study. In *2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 206–214, 2022. doi: 10.1109/ISMAR55827.2022.00035 7

[24] A. Gentile, S. Ficarra, E. Thomas, A. Bianco, and A. Nordstrom. Nature through virtual reality as a stress-reduction tool: A systematic review. *Int. J. Stress Manag.*, 30(4):341–353, Nov. 2023. 2

[25] S. M. Gerber, M.-M. Jeitziner, P. Wyss, A. Chesham, P. Urwyler, R. M. Müri, S. M. Jakob, and T. Nef. Visuo-acoustic stimulation that helps you to relax: A virtual reality setup for patients in the intensive care unit. *Sci. Rep.*, 7(1):13228, Oct. 2017. 7

[26] K. Guidolin, F. Jung, S. Hunter, H. Yan, M. Englesakis, S. Verderber, S. Chadi, and F. Quereshy. The influence of exposure to nature on inpatient hospital stays: A scoping review. *HERD*, 17(2):360–375, Apr. 2024. 2

[27] J. P. Guilford. *The Nature of Human Intelligence*. McGraw-Hill, New York, 1967. 7

[28] S. G. Hart and L. E. Staveland. Development of nasa-tlx (task load index): Results of empirical and theoretical research. In P. A. Hancock and N. Meshkati, eds., *Human Mental Workload*, vol. 52 of *Advances in Psychology*, pp. 139–183. North-Holland, 1988. doi: 10.1016/S0166-4115(08)62386-9 7

[29] T. Hartig. Validation of a measure of perceived environmental restorativeness. *Goteborg psychological reports*, 26(7), 1996. 6

[30] T. Hartig, R. Mitchell, S. De Vries, and H. Frumkin. Nature and health. *Annual review of public health*, 35(1):207–228, 2014. doi: 10.1146/annurev-publhealth-032013-182443 1

[31] F. Hesam Shariati, A. Steffens, and S. Adhami. Designing environments that contribute to a reduction in the progression of parkinson's disease; a literature review. *Health & Place*, 83:103105, 2023. doi: 10.1016/j.healthplace.2023.103105 2

[32] S.-H. Jo, J.-S. Park, and P.-S. Yeon. The effect of forest video using virtual reality on the stress reduction of university students focused on C university in korea. *Int. J. Environ. Res. Public Health*, 18(23):12805, Dec. 2021. 6

[33] D. Jung, D. I. Kim, and N. Kim. Bringing nature into hospital architecture: Machine learning-based eeg analysis of the biophilia effect in virtual reality. *Journal of environmental psychology*, 89:102033–, 2023. 5

[34] S. Kaplan. The restorative benefits of nature: Toward an integrative framework. *Journal of Environmental Psychology*, 15(3):169–182, 1995. Green Psychology. doi: 10.1016/0272-4944(95)90001-2 2, 6, 7

[35] H. Kim, D. J. Kim, S. Kim, W. H. Chung, K.-A. Park, J. D. K. Kim, D. Kim, M. J. Kim, K. Kim, and H. J. Jeon. Effect of virtual reality on stress reduction and change of physiological parameters including heart rate variability in people with high stress: An open randomized crossover trial. *Front. Psychiatry*, 12:614539, Aug. 2021. 6

[36] I. Ladakis, D. Filos, and I. Chouvarda. Virtual reality environments for stress reduction and management: a scoping review. *Virtual Reality*, 28(1):50, 2024. 2

[37] A. Latini, S. Torresin, T. Oberman, E. Di Giuseppe, F. Aletta, J. Kang, and M. D'Orazio. Virtual reality application to explore indoor soundscape and physiological responses to audio-visual biophilic design interventions: An experimental study in an office environment. *Journal of Building Engineering*, 87:108947, 2024. doi: 10.1016/j.jobe.2024.108947 6

[38] M. Lee, E. Kim, J. Choe, S. Choi, S. Ha, and G. Kim. Psychological effects of green experiences in a virtual environment: A systematic review. *Forests*, 13(10), 2022. doi: 10.3390/f13101625 2

[39] G. Y. Leung, H. Hazan, and C. S. Chan. Exposure to nature in immersive virtual reality increases connectedness to nature among people with low nature affinity. *Journal of environmental psychology*, 83:101863–, 2022. 7

[40] J. Li, Y. Ma, P. Li, and A. Butz. A journey through nature: Exploring virtual restorative environments as a means to relax in confined spaces. *A Journey Through Nature*, article no. 22, 9 pages, 2021. doi: 10.1145/3450741.3465248 7

[41] Z. Li, Y. Wang, H. Liu, and H. Liu. Physiological and psychological effects of exposure to different types and numbers of biophilic vegetable walls in small spaces. *Building and Environment*, 225:109645, 2022. doi: 10.1016/j.buildenv.2022.109645 5

[42] S. Litleskare, F. Fröhlich, O. E. Flaten, A. Haile, S. Å. Kjøs Johnsen, and G. Calogiuri. Taking real steps in virtual nature: a randomized blinded trial. *Virtual reality*, 26(4):1777–1793, 2022. doi: 10.1007/s10055-022-00670-21

[43] P. Liu, J. Liu, J. Fernandez, Q. Zuo, and M. Lin. Positive affect and natural landscape in virtual reality: A systematic review comparing interventions, measures, and outcomes. *Journal of Environmental Psychology*, 88:102011, 2023. doi: 10.1016/j.jenvp.2023.102011 2

[44] R. Lundstedt, C. Håkansson, M. Lõhmus, and M. Wallergård. Designing virtual natural environments for older adults in residential care facilities. *Technology and Disability*, 33(4):305–318, 2021. doi: 10.3233/TAD-210344 5

[45] K. Lyu, R. de Dear, A. Brambilla, and A. Globa. Restorative benefits of semi-outdoor environments at the workplace: Does the thermal realm matter? *Building and Environment*, 222:109355, 2022. doi: 10.1016/j.buildenv.2022.109355 6

[46] K. Lyu, R. Dear, A. Brambilla, and A. Globa. Restorative experience in semi-outdoor spaces: from thermal pleasure to psychological well-being. 07 2024. 6

[47] A. Mahrous, K. Dewidar, M. Refaat, and A. Nessim. The impact of biophilic attributes on university students level of satisfaction: Using virtual reality simulation. *Ain Shams Engineering Journal*, 15(1):102304, 2024. doi: 10.1016/j.asej.2023.102304 5

[48] R. Masters, V. Interrante, M. Watts, and F. Ortega. Virtual nature: Investigating the effect of biomass on immersive virtual reality forest bathing applications for stress reduction. In *ACM Symposium on Applied Perception 2022*, SAP '22, article no. 6, 10 pages. Association for Computing Machinery, New York, NY, USA, 2022. doi: 10.1145/3548814.3551459 6

[49] R. Masters, J. Nicoly, V. Gaddy, V. Interrante, and F. Ortega. The impact of nature realism on the restorative quality of virtual reality forest bathing. *ACM Trans. Appl. Percept.*, 22(1), article no. 3, 18 pages, Nov. 2024. doi: 10.1145/3670406 6

[50] O. Mattila, A. Korhonen, E. Pöyry, K. Hauru, J. Holopainen, and P. Parvinen. Restoration in a virtual reality forest environment. *Computers in Human Behavior*, 107:106295, 2020. doi: 10.1016/j.chb.2020.106295 6

[51] F. S. Mayer and C. M. Frantz. The connectedness to nature scale: A measure of individuals' feeling in community with nature. *Journal of environmental psychology*, 24(4):503–515, 2004. 7

[52] N. McKeever, A. Nezami, and D. Kourtis. The overview effect and nature-relatedness. *Frontiers in Virtual Reality*, 5, 2024. doi: 10.3389/frvir.2024.1196312 8

[53] M. Melo, G. Gonçalves, P. Monteiro, H. Coelho, J. Vasconcelos-Raposo, and M. Bessa. Do multisensory stimuli benefit the virtual reality experience? a systematic review. *IEEE Transactions on Visualization and Computer Graphics*, 28(2):1428–1442, 2022. doi: 10.1109/TVCG.2020.3010088 8

[54] S. Meshkat, M. Edalatkhah, C. Di Luciano, J. Martin, G. Kaur, G. H. Lee, H. Park, A. Torres, A. Mazalek, B. Kapralos, et al. Virtual reality and stress management: A systematic review. *Cureus*, 16(7), 2024. 2

[55] M. Mollazadeh and Y. Zhu. Application of virtual environments for biophilic design: A critical review. *Buildings*, 11(4), 2021. doi: 10.3390/buildings11040148 2

[56] F. Mostajeran, M. Fischer, F. Steinicke, and S. Kühn. Effects of exposure to immersive computer-generated virtual nature and control environments on affect and cognition. *Sci Rep*, 13(1):220, Jan. 2023. 5

[57] F. Mostajeran, S. Schneider, G. Bruder, S. Kühn, and F. Steinicke. Analyzing cognitive demands and detection thresholds for redirected walking in immersive forest and urban environments. *Analyzing Cognitive Demands and Detection Thresholds for Redirected Walking in Immersive Forest*

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and Urban Environments, pp. 61–71, 2024. doi: 10.1109/VR58804.2024.00030 7

[58] J. Navarrete, J. Navarro-Siurana, R. Herrero, M. D. Vara, M. Miragall, and R. Baños. Connecting to nature through 360° videos during COVID-19 confinement: A pilot study of a brief psychological intervention. *J. Healthc. Eng.*, 2022:4242888, Sept. 2022. 5

[59] J. B. Nicoly, R. Masters, V. Gaddy, V. Interrante, and F. Ortega. The restorative influence of virtual reality environment design. In *ACM Symposium on Applied Perception 2024*, SAP '24, article no. 12, 10 pages. Association for Computing Machinery, New York, NY, USA, 2024. doi: 10.1145/3675231.3675244 6

[60] E. K. Nisbet, J. M. Zelenski, and S. A. Murphy. The nature relatedness scale: Linking individuals' connection with nature to environmental concern and behavior. *Environment and behavior*, 41(5):715–740, 2009. 7

[61] F. R. Ortega, V. Interrante, S. Lotemplio, R. Masters, J. Nicoly, Z. Borhani, D. Davalos, and D. Zielasko. Enhancing Well-Being Through Positive Technology: VR Forest Bathing, 2024. 1

[62] M. J. Page et al. The prisma 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, 372, 2021. doi: 10.1136/bmj.n71 2

[63] A. Palanica, A. Lyons, M. Cooper, A. Lee, and Y. Fossat. A comparison of nature and urban environments on creative thinking across different levels of reality. *Journal of Environmental Psychology*, 63:44–51, 2019. doi: 10.1016/j.jenvp.2019.04.006 7

[64] G. Reese, J. Stahlberg, and C. Menzel. Digital shinrin-yoku: do nature experiences in virtual reality reduce stress and increase well-being as strongly as similar experiences in a physical forest? *Virtual Reality*, 26:1–11, 02 2022. doi: 10.1007/s10055-022-00631-9 6

[65] W. Ren, Y. Gu, and Y. Tao. Research on the methods of vr technology to reduce the stress of college students: A case-study of virtual aquarium. In *Proceedings of the 5th International Conference on Digital Technology in Education*, ICDTE '21, 6 pages, p. 124–129. Association for Computing Machinery, New York, NY, USA, 2022. doi: 10.1145/3488466.3488481 6

[66] S. Romano, E. Laviola, A. E. Uva, and M. Gattullo. Biophilic Design of Virtual Workplaces: Effect of Animations on User Attention. In *Virtual Reality and Mixed Reality: 20th EuroXR International Conference, EuroXR 2023, Rotterdam, The Netherlands, November 29 – December 1, 2023, Proceedings*, p. 201–214. Springer-Verlag, Berlin, Heidelberg, 2023. doi: 10.1007/978-3-031-48495-7_14 7

[67] I. Sadowski and B. Khoury. Nature-based mindfulness-compassion programs using virtual reality for older adults: A narrative literature review. *Frontiers in Virtual Reality*, 3, 2022. doi: 10.3389/frvir.2022.892905 2

[68] E. Sansal, A. Candan Simsek, S. Aktan, F. Özbey, and A. Paksoy. Restorative effects of virtual nature on the emotional well-being of community-dwelling older adults. *European Journal of Geriatrics and Gerontology*, 6:12–18, 05 2024. doi: 10.4274/ejgg.galenos.2023.2023-3-2 7

[69] D. Scates, J. I. Dickinson, K. Sullivan, H. Cline, and R. Balaraman. Using nature-inspired virtual reality as a distraction to reduce stress and pain among cancer patients. *Environment and Behavior*, 52(8):895–918, 2020. doi: 10.1177/0013916520916259 6

[70] P. W. Schultz. *Inclusion with Nature: The Psychology Of Human-Nature Relations*, pp. 61–78. Springer US, Boston, MA, 2002. doi: 10.1007/978-1-4615-0995-0_4 7

[71] L. Sharam, K. Mayer, and O. Baumann. Design by nature: The influence of windows on cognitive performance and affect. *Journal of environmental psychology*, 85:101923–, 2023. 7

[72] D. Sheffield, C. Butler, and M. Richardson. Improving nature connectedness in adults: A meta-analysis, review and agenda. *Sustainability*, 14:12494, 09 2022. doi: 10.3390/su141912494 2

[73] M. D. Smith, S. Getchell, and M. Weatherly. Human connectedness to nature: Comparison of natural vs. virtual experiences. In T.-T. Wu, Y.-M. Huang, R. Shadiev, L. Lin, and A. I. Starčič, eds., *Innovative Technologies and Learning*, pp. 215–219. Springer International Publishing, Cham, 2018. 8

[74] G. Spano, A. Theodorou, G. Reese, G. Carrus, G. Sanesi, and A. Panno. Virtual nature and psychological and psychophysiological outcomes: A systematic review. *Journal of environmental psychology*, 89:102044–, 2023. 2

[75] C. D. Spielberger, R. Goruch, R. Lushene, P. Vagg, and G. Jacobs. Manual for the state-trait inventory stai (form y). *Mind Garden, Palo Alto, CA, USA*, 1983. 5, 6

[76] V. Spors, S. Laato, O. O. Buruk, and J. Hamari. Longing to be the mountain: A scoping review about nature-centric, health-minded technologies. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI '23, article no. 523, 16 pages. Association for Computing Machinery, New York, NY, USA, 2023. doi: 10.1145/3544548.3581479 2

[77] A. Tural and E. Tural. Comparative analysis of restorative interior design elements: Screen-based versus virtual reality evaluations for future medical treatment prospects. *Int. J. Environ. Res. Public Health*, 22(1):44, Dec. 2024. 3, 6

[78] R. S. Ulrich, R. F. Simons, B. D. Losito, E. Fiorito, M. A. Miles, and M. Zelson. Stress recovery during exposure to natural and urban environments. *Journal of Environmental Psychology*, 11(3):201–230, 1991. doi: 10.1016/S0272-4944(05)80184-7 6

[79] L. Verzwyvelt, A. McNamara, X. Xu, and R. Stubbins. Effects of virtual reality v. biophilic environments on pain and distress in oncology patients: a case-crossover pilot study. *Scientific Reports*, 11:20196, 10 2021. doi: 10.1038/s41598-021-99763-2 5

[80] D. Watson, L. A. Clark, and A. Tellegen. Development and validation of brief measures of positive and negative affect: the panas scales. *Journal of personality and social psychology*, 54(6):1063, 1988. 5

[81] S. Webber, R. M. Kelly, G. Wadley, and W. Smith. Engaging with nature through technology: A scoping review of hci research. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI '23, article no. 521, 18 pages. Association for Computing Machinery, New York, NY, USA, 2023. doi: 10.1145/3544548.3581534 2

[82] V. S. Williams, R. J. Morlock, and D. Feltner. Psychometric evaluation of a visual analog scale for the assessment of anxiety. *Health and quality of life outcomes*, 8:1–8, 2010. 6

[83] E. O. Wilson. *Biophilia*. Harvard University Press, Cambridge, MA, 1984. 2

[84] A. Winn, A. Jayadas, T. Chandrasekera, and S. Thaxton. Biophilic interventions in space habitat crew quarters to improve cognitive & physiological health. In *2023 IEEE Aerospace Conference*, pp. 1–14, 2023. doi: 10.1109/AERO55745.2023.10115991 7

[85] S. Xiaoxue and X. Huang. Promoting stress and anxiety recovery in older adults: assessing the therapeutic influence of biophilic green walls and outdoor view. *Frontiers in Public Health*, 12, 2024. doi: 10.3389/fpubh.2024.1352611 6

[86] N. Yeo, M. White, I. Alcock, R. Garside, S. Dean, A. Smalley, and B. Gatersleben. What is the best way of delivering virtual nature for improving mood? an experimental comparison of high definition tv, 360° video, and computer generated virtual reality. *Journal of Environmental Psychology*, 72:101500, 2020. doi: 10.1016/j.jenvp.2020.101500 8

[87] S. Yeom, H. Kim, and T. Hong. Psychological and physiological effects of a green wall on occupants: A cross-over study in virtual reality. *Building and Environment*, 204:108134, 2021. doi: 10.1016/j.buildenv.2021.108134 6

[88] J. Yin, N. Arfaei, P. MacNaughton, P. J. Catalano, J. G. Allen, and J. D. Spengler. Effects of biophilic interventions in office on stress reaction and cognitive function: A randomized crossover study in virtual reality. *Indoor Air*, 29(6):1028–1039, Nov. 2019. 6

[89] J. Yin, J. Yuan, N. Arfaei, P. J. Catalano, J. G. Allen, and J. D. Spengler. Effects of biophilic indoor environment on stress and anxiety recovery: A between-subjects experiment in virtual reality. *Environment International*, 136:105427, 2020. doi: 10.1016/j.envint.2019.105427 6

[90] J. Yin, S. Zhu, P. MacNaughton, J. G. Allen, and J. D. Spengler. Physiological and cognitive performance of exposure to biophilic indoor environment. *Building and Environment*, 132:255–262, 2018. doi: 10.1016/j.buildenv.2018.01.006 7

[91] H. Yoon and J. Y. Jeon. Relationship between soundscape perception and psychophysiological responses to virtual environmental events. *Virtual Reality*, 29, 12 2024. doi: 10.1007/s10055-024-01087-9 6

[92] J. You, X. Wen, L. Liu, J. Yin, and J. S. Ji. Biophilic classroom environments on stress and cognitive performance: A randomized crossover study in virtual reality (VR). *PLoS One*, 18(11):e0291355, Nov. 2023. 7, 8

[93] C.-P. Yu, H.-Y. Lee, and X.-Y. Luo. The effect of virtual reality forest and urban environments on physiological and psychological responses. *Urban Forestry & Urban Greening*, 35:106–114, 2018. doi: 10.1016/j.ufug.2018.08.013 6

[94] G. Zhang, G. Wu, and J. Yang. The restorative effects of short-term exposure to nature in immersive virtual environments (ives) as evidenced by participants' brain activities. *Journal of Environmental Management*, 326:116830, 2023. doi: 10.1016/j.jenvman.2022.116830 6

[95] M. Zuckerman. Development of a situation-specific trait-state test for the prediction and measurement of affective responses. *Journal of Consulting and Clinical Psychology*, 1977. 5

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